

ORACLE EPM PREDICTIVE CASH FORECASTING

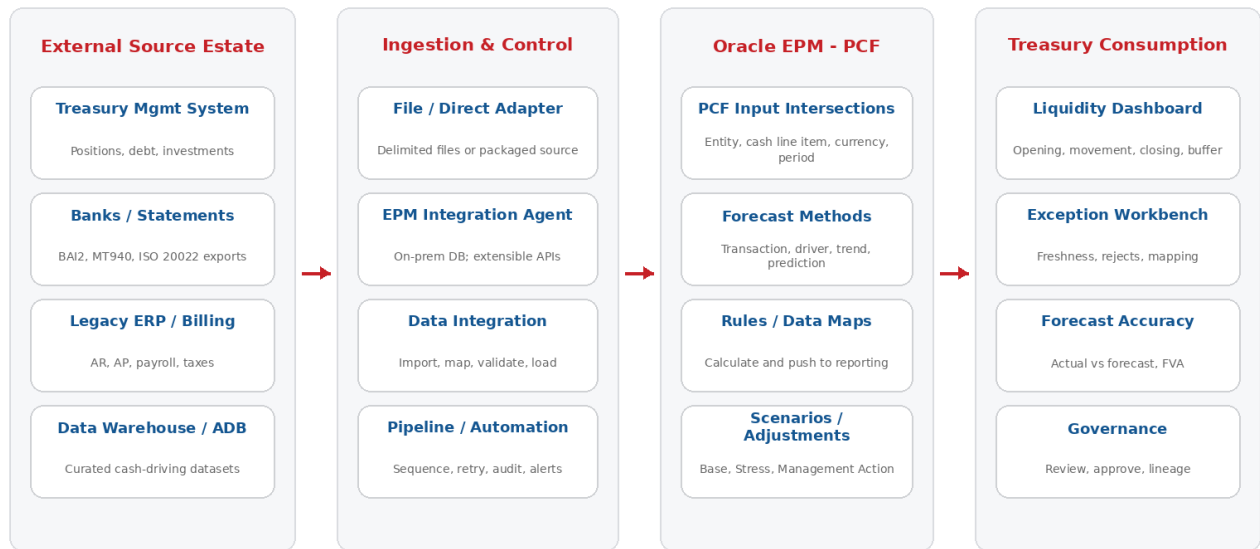
External Data Sources

Advanced Lab Guide | Product Design + Implementation + Training + Technical Reference + Troubleshooting

Advanced-stage outcome

Design a production-grade external-data integration framework for Predictive Cash Forecasting that ingests non-Fusion data at the correct cash-forecast granularity, preserves lineage, supports restart and reconciliation, and gives treasury users reliable inputs for forecasting and liquidity decisions.

External Data Sources Reference Architecture



1. Guide Purpose, Audience and Outcomes

This guide is intended for advanced Oracle EPM consultants, treasury solution architects, integration developers, administrators, support teams and senior finance users. It treats external data ingestion as a governed product capability - not merely a file load - covering source contracts, integration patterns, mapping, time granularity, controls, lineage, automation, security, forecasting and incident recovery.

By the end of the lab, students will be able to

- Explain how external/non-Fusion data feeds Predictive Cash Forecasting and where Data Integration fits.
- Choose among file-based integration, packaged/direct adapters and EPM Integration Agent patterns.
- Use PCF data-load templates as a target-intersection reference while implementing production loads through Data Integration.
- Align daily, weekly or monthly source aggregation to the cash-forecast granularity configured in PCF.
- Map Entity, Cash Flow Line Item, Currency, Scenario, Period and custom dimensions without losing traceability.
- Build source-to-load reconciliation, freshness controls, idempotent reruns and exception blocking.
- Integrate external TMS/bank/legacy data into a 13-week forecast and assess liquidity impact.
- Troubleshoot file, agent, API, period, mapping, sign, security and data-quality failures.

Document modes

Use	What this guide provides
Product design documentation	Reference architecture, UI concepts, source contract, lineage and control framework.
Implementation guide	Source registration, import format, period mapping, member mappings, load options and orchestration.
Training material	Hands-on 13-week lab with realistic source datasets and expected calculations.
Technical reference	Granularity rules, mappings, source types, Integration Agent pattern and run metadata.
Troubleshooting handbook	Decision tree, log examples and symptom/root-cause/remediation matrix.

2. Oracle Capability Baseline for External Sources

Current Oracle guidance

Predictive Cash Forecasting can pull data from external systems, and Oracle recommends Data Integration as the preferred integration mechanism for external sources. PCF provides sample data-load templates as guidance for the target intersections expected by the application. For non-Fusion ERP data, data must be aggregated at the daily, weekly or monthly granularity selected when cash forecasting features were enabled.

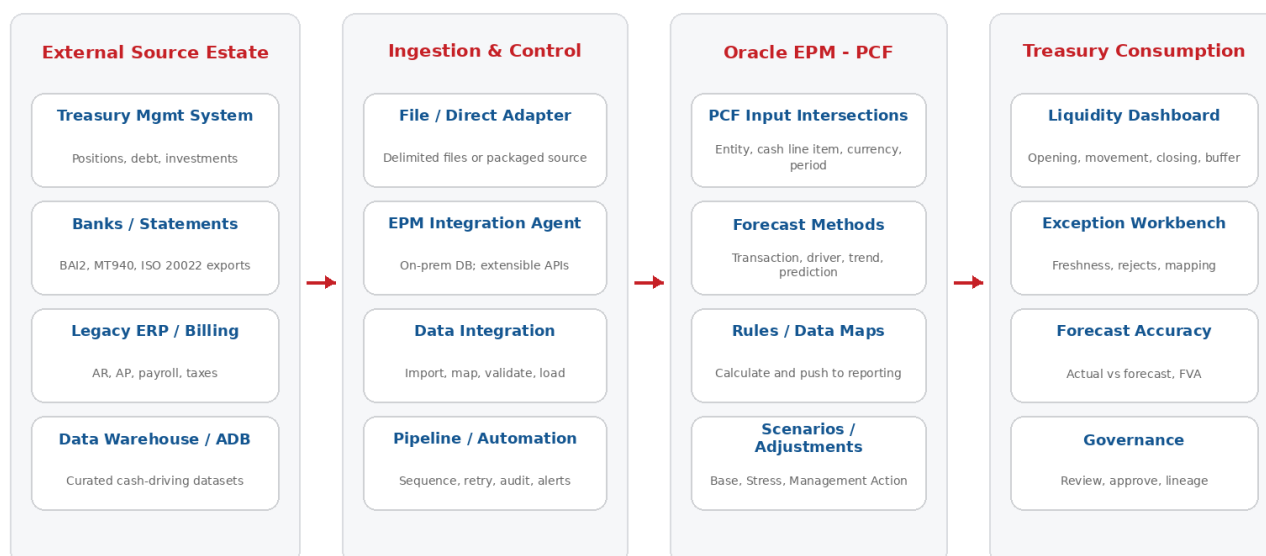
Capability	Implementation meaning
External-system data	Outstanding transactions, balances, drivers and forecast inputs can be loaded from non-Fusion sources.
Data Integration preferred	Use Data Integration for governed production mappings, period handling, load rules and operations.
Sample load templates	Use generated PCF templates to understand expected intersections and for controlled/manual testing.
Granularity alignment	External data must align to the application time grain: daily, weekly or monthly.
File-based sources	Delimited/fixed-width source files can be registered and mapped to an EPM target.
Direct/packaged sources	Supported Oracle/enterprise source adapters can be used where available.
EPM Integration Agent	Supports on-prem relational extraction and can be extended through scripts for APIs/non-relational sources.

Design implication

An external-source program should standardize data before it reaches the forecast model. PCF should receive decision-ready cash data at the configured time grain, with explicit source timestamp, entity, currency, cash-flow line item and scenario semantics.

3. Reference Architecture

External Data Sources Reference Architecture



Architecture principles

Principle	Design rule
Source contract	Every source owns a schema, grain, cut-off time, sign convention and reconciliation total.
Least transformation	Transform only what is needed to express source data in PCF target dimensions.
Traceability	Retain source file/job ID, timestamp and row/business key outside or alongside target load evidence.
Idempotency	A rerun of the same source slice must not double count cash.
Granularity control	Aggregate upstream to the PCF daily/weekly/monthly grain before or during integration.
Separation	Operational/TMS systems own source transactions; PCF owns forecast methods, scenarios and planning adjustments.
Observability	Capture row counts, rejected rows, source/target totals, freshness and final status.

4. External Source Portfolio and Cash Use Cases

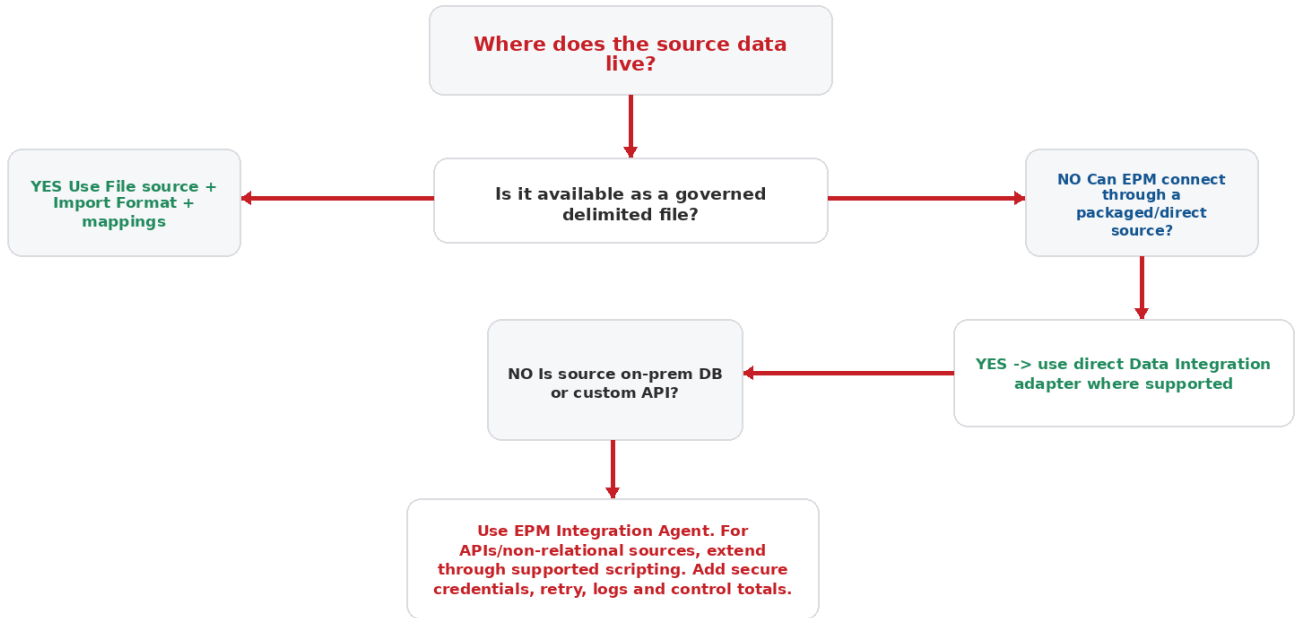
Source	Typical data	PCF usage	Integration pattern
Treasury Management System	Debt, investments, liquidity positions, intercompany funding	Financing/investing cash flows; opening position	File, database/agent or API pattern
Bank / Statement feed	Balances and cash movements	Opening cash, actual cash movement	Bank export file or upstream staging
Legacy ERP / Billing	Customer/supplier invoices, payroll, taxes	Operating inflows/outflows	File, DB agent, warehouse extract
Data Warehouse / Lakehouse	Curated cash drivers and history	Drivers, actual history, reconciliation	Direct supported source or Integration Agent
Oracle Autonomous Database	Staged operational/analytics data	Curated source for PCF	Data Integration / Integration Agent pattern
Third-party API	Macro rates, TMS positions, payment platform data	External drivers or cash events	Integration Agent extension / controlled staging
Spreadsheet / CSV	Approved manual source feed	Low-volume controlled input	File-based Data Integration

Avoid these anti-patterns

- Loading transaction-level detail into a monthly PCF model when only monthly aggregates are required.
- Allowing each source team to invent its own entity, currency and sign convention.
- Using spreadsheets as permanent production interfaces without ownership, versioning and control totals.
- Mixing source cleansing, forecast judgment and business overrides in the same integration step.
- Relying on file names alone instead of run IDs, timestamps and checksums for lineage.

5. Integration Pattern Decision Tree

Choose the External Integration Pattern



Pattern selection summary

Pattern	Best fit	Key control
File source	Governed CSV/delimited export from external system	Schema validation, checksum, file timestamp and duplicate detection
Direct / packaged adapter	Supported source with native Data Integration connectivity	Connection security, source filter and period mapping
EPM Integration Agent	On-prem relational DB or custom source needing controlled extraction	Agent cluster health, SQL/script version, credentials, retry
API -> governed staging	Third-party API not directly modeled as Data Integration source	Stage response, normalize schema, then load through governed integration

6. Data Contract and Target Intersection Design

Field	Example	Control
Source_System	TMS01	Registered source ID; immutable for lineage
Run_ID	TMS_20260826_0630	Unique execution key
Source_Timestamp	2026-08-26 06:15 IST	Freshness threshold check
Entity	US_TREASURY_01	Maps to PCF Entity
Cash_Flow_Line	Debt Repayment	Maps to PCF cash line item
Currency	USD	Valid PCF currency/member
Period	2026-W35	Maps to target weekly period
Scenario	Working	Controlled category/scenario mapping
Amount	-1,250,000	Sign convention validated
Source_Key	TERMLOAN-A-2026W35	Duplicate/reconciliation key

Granularity rule

For a weekly PCF application, a source may arrive as detailed daily transactions, but the governed load should aggregate them into the applicable weekly target intersection. For a daily application, preserve daily grain; for monthly, load monthly aggregates.

7. Oracle-Inspired Data Integration Configuration

Oracle-inspired | Data Integration

Data Exchange > Data Integration

Create Integration - External Treasury Source

General

Map Dimensions

Map Members

Options

Filters

Run Integration

Process Details

Integration Definition

Name: PCF_TMS_13W_CASH

Location: PCF_TMS_EXTERNAL

Source: TMS_Cash_File

Target: Cash Forecasting

Category: Actual / Working

Period Mapping: Weekly - ISO week aligned

Control Options

- ✓ Import Mode: Replace
- ✓ Export Mode: Store Data
- ✓ Check Period Mapping: Yes
- ✓ Reject unmapped: Yes
- ✓ Source timestamp required: Yes

Target POV

- Entity: US_TREASURY_01
- Currency: USD
- Scenario: Working
- Cash Flow: External Source
- Week: W01-W13
- Data Source: TMS

Save and Run

Core setup objects

- Source application: File, supported direct source, or Integration Agent-backed source.
- Target application: Predictive Cash Forecasting application.
- Location: binds source, target and member mappings; use location security for controlled access.
- Import Format / column mappings: maps source fields to target dimensions.
- Period mappings: explicit calendar-to-PCF mapping; test W53 and fiscal calendar exceptions.
- Category mapping: maps integration category to the target Scenario member.
- Member mappings: explicit/range/like/multi-dimension logic as appropriate.
- Options: import/export mode, filters, target slice, validation and workflow behavior.

8. Sample Data Load Templates - Correct Use

PCF can generate sample data import templates based on enabled features and custom dimensions. Use them to understand expected target intersections and to perform controlled testing. Oracle recommends Data Integration as the preferred production method for integrating external sources.

Lab procedure

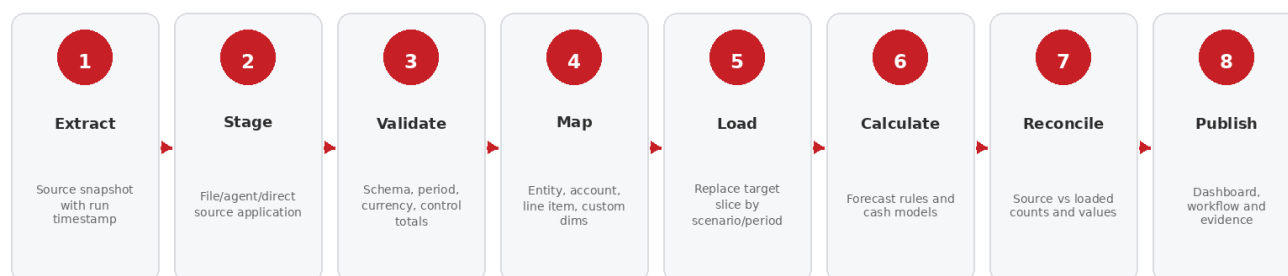
1. In PCF, navigate to Application > Configure.
2. From Actions, select Download Data Load Templates.
3. Save and extract the generated ZIP.
4. Review the CSV structure for the enabled daily/periodic model and custom dimensions.
5. Identify the target dimensions required for your external source.
6. Create a Data Integration source/import format matching your governed source layout.
7. Map source columns/members to the PCF target.
8. Run a small controlled load and compare the target result to the template-based expectation.

Training vs production

Direct template import is useful for learning, prototyping and controlled recovery. For recurring production loads, prefer governed Data Integration rules with mappings, logs and reconciliation evidence.

9. End-to-End Process Flow

External Source -> PCF Controlled Load Cycle

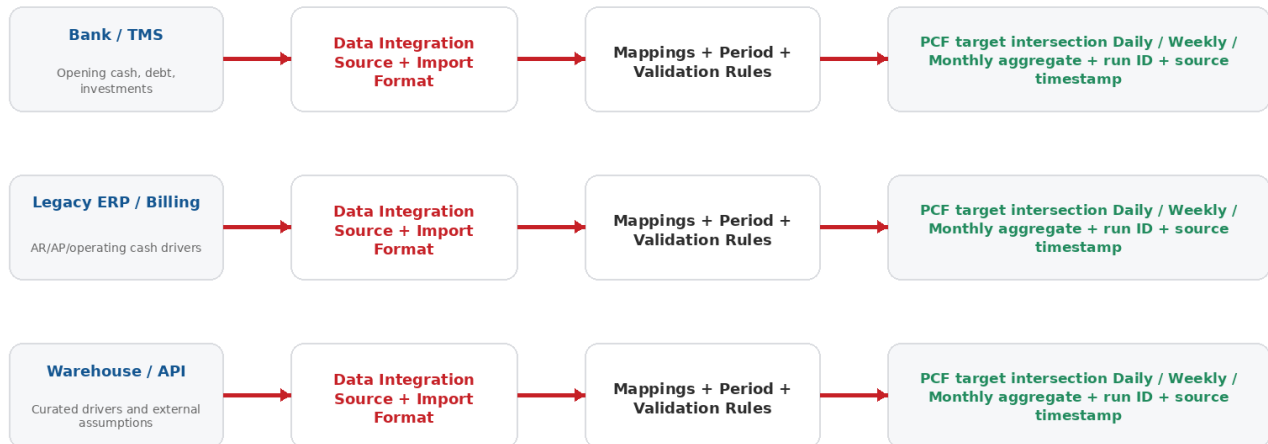


Stage gates

Stage	Pass condition
Extract	Source is complete, timestamped, checksum/row count captured.
Stage	Correct source application and file/query version used.
Validate	Required columns, dates, periods, currencies and numeric amounts valid.
Map	No critical unmapped Entity/Line Item/Currency/Scenario members.
Load	Target slice loaded once using controlled replace/merge semantics.
Calculate	PCF rules complete with no blocking errors.
Reconcile	Source total equals accepted target total after documented exclusions.
Publish	Critical exceptions = 0 and reviewer signs off.

10. Data Flow, Granularity and Lineage

Data Flow, Granularity and Lineage



Non-Fusion ERP source data should be aggregated to the PCF time granularity enabled in the application.

Lineage evidence to retain

- Source system and source object/file name.
- Source extraction timestamp and business cut-off.
- Run ID, integration/location and job ID.
- Source row count and source control total.
- Rejected count and value, with reason codes.
- Target accepted row count and amount.
- Period mapping and member mapping version.
- User/service account executing the load.
- Forecast rule/calculation job executed after load.

11. Mapping Framework for External Cash Sources

Source value	Target example	Mapping type	Reason
LEGAL_US01	US_TREASURY_01	Explicit	Legal entity crosswalk
BANK_*_USD	Cash Management / USD	Like/wildcard	Bank cash grouping
DEBT_TERM_*	Debt Repayments	Like/wildcard	Financing cash-flow grouping
CAPEX_1000-1999	Capital Expenditure	Range	Project/capex codes
PAYROLL US REG	Payroll	Multi-dimension	Combination-based cash semantics
Other / unmapped	Reject - do not default	Validation rule	Avoid silent misclassification

Mapping governance

- Do not map unknown members to a generic “Other” if the amount could be material.
- Version mappings and record effective dates where source master data changes.
- Separate technical normalization (trim/case) from business classification.
- Test negative/positive signs for inflows/outflows after mapping, not before only.

12. Period Mapping and Calendar Edge Cases

Case	Risk	Control
ISO week vs fiscal week	Cash falls into wrong forecast bucket	Explicit source period mapping and UAT calendar crosswalk
Week 53	Rejected or shifted values	Define W53 behavior before production
Month-end spanning week	Double count between weekly/monthly feeds	Single authoritative grain per enabled PCF model
Time zone cut-off	Late file appears current but contains prior-day data	Persist source business timestamp, not upload timestamp
Holiday calendar	Expected settlement shifts	Handle in source timing/forecast method, not arbitrary period remap

13. Advanced Lab Scenario - 13-Week External Source Integration

BISP Treasury is implementing a 13-week PCF forecast. Fusion ERP provides core AR/AP, while three external feeds are added: a TMS debt/investment feed, a legacy payroll/tax feed and an external bank-position feed. The objective is to integrate external cash events, calculate closing cash and detect a funding gap.

Source	External item	W1	W2	W3	W4	W5	W6
TMS	Debt repayment	0	0	(1.25)	0	0	(0.75)
TMS	Interest payment	(0.18)	0	0	(0.20)	0	0
Legacy	Payroll	(1.10)	(1.10)	(1.10)	(1.10)	(1.10)	(1.10)
Legacy	Tax payment	0	0	(0.85)	0	0	0
Bank	Opening cash	8.00					

Amounts are USD millions. The core operating forecast before these external feeds produces weekly net operating cash movements of +0.30, +0.45, +0.20, +0.55, +0.40 and +0.35 for W1-W6.

14. Lab Step 1 - Prepare and Validate External Files

1. Create TMS, Legacy Payroll and Bank source files with the data contract fields defined earlier.
2. Assign a unique Run_ID and Source_Timestamp to each file.
3. Confirm amount sign convention: receipts positive, payments negative.
4. Validate all Entity, Currency and Period values against PCF metadata.
5. Calculate source control totals before loading.
6. Simulate one invalid currency and one stale bank timestamp to test exception handling.

Source	Rows	Control total (USD M)	Freshness SLA	Expected status
TMS	14	-2.38	<= 2 hours	PASS
Legacy Payroll/Tax	19	-7.45	<= 4 hours	PASS
Bank Position	6	8.00 opening balance	<= 1 hour	PASS

15. Lab Step 2 - Configure Data Integrations

1. Register each source application or agent-backed source.
2. Create separate locations PCF_TMS_EXTERNAL, PCF_LEGACY_PAYROLL and PCF_BANK_EXTERNAL.
3. Define import formats for source columns and target dimensions.
4. Create weekly source period mappings for 2026-W35 onward.
5. Map Entity, Cash Flow Line Item, Currency, Scenario and Data Source.
6. Set controlled import/export mode so reruns replace the intended slice rather than append duplicate values.
7. Run mappings validation and resolve all critical unmapped members before production load.

Advanced implementation note

A separate location per governed source improves ownership, mapping isolation, location security and restartability. Do not over-consolidate unrelated external feeds into one opaque integration.

16. Lab Step 3 - Load, Calculate and Reconcile

1. Run the Bank integration first so opening cash is current.
2. Run TMS and Legacy integrations and capture job IDs.
3. Confirm source rows, accepted rows and rejected rows.
4. Run the PCF cash forecasting model/rules in the required sequence.
5. Push to the reporting cube/data map if used in your reporting design.
6. Reconcile source totals to loaded PCF external-source values.
7. Block publication if any critical mapping or freshness exception remains.

Control	Expected result
Bank opening cash	8.00M
TMS loaded total W1-W6	-2.38M
Legacy payroll/tax W1-W6	-7.45M
Critical rejects	0 after remediation
Source-to-load difference	0.00M
Run status	SUCCESS

17. Lab Step 4 - Calculate Closing Cash and Liquidity Impact

Week	Opening	Core operating net	External TMS	Legacy payroll/tax	Closing
W1	8.00	0.30	-0.18	-1.10	7.02
W2	7.02	0.45	0.00	-1.10	6.37
W3	6.37	0.20	-1.25	-1.95	3.37
W4	3.37	0.55	-0.20	-1.10	2.62
W5	2.62	0.40	0.00	-1.10	1.92
W6	1.92	0.35	-0.75	-1.10	0.42

Decision point

Assume the minimum liquidity buffer is USD 1.50M. W6 closing cash falls to USD 0.42M, creating a USD 1.08M shortfall. The treasury action is not an integration correction: the integration has exposed a real financing need. Create a Management Action scenario for a drawdown or payment deferral.

18. Scenario Stress and External Assumption Feed

Extend the lab by loading an external macro assumption feed that increases interest by 100 bps and delays one investment maturity by two weeks. Use the feed as a controlled external driver, not a manual overwrite of source actuals.

Scenario	Change	W6 closing cash	Buffer gap
Base	As loaded	0.42M	-1.08M
Stress	Higher interest + delayed maturity	0.05M	-1.45M
Management Action	1.75M revolver draw in W5	1.80M	+0.30M

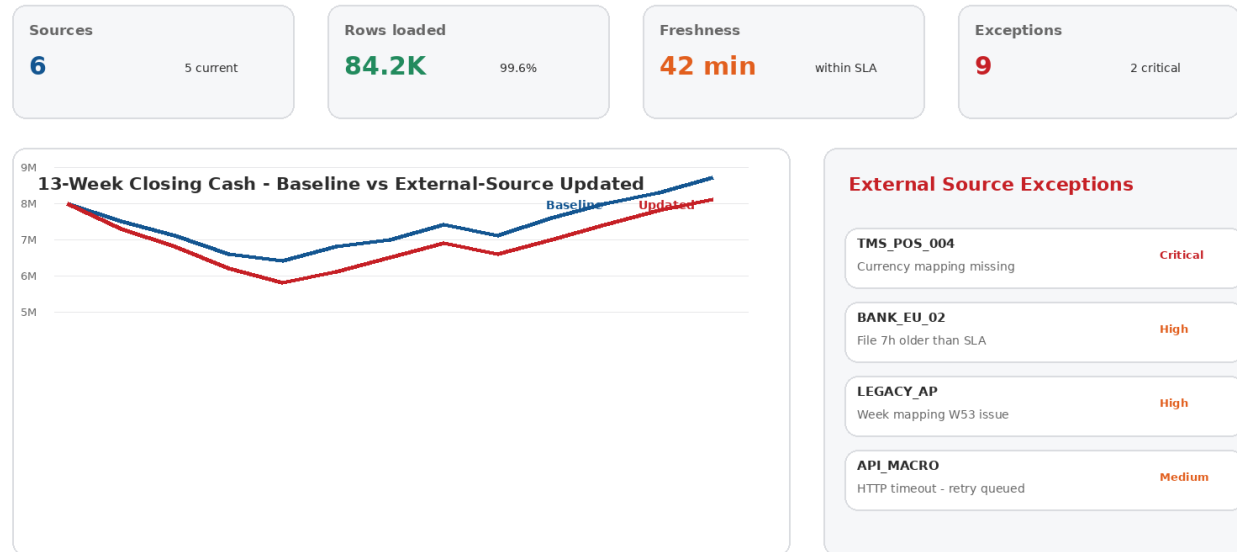
Student task

- Identify which values belong to source actual/external data versus planning scenario assumptions.
- Explain why a macro API failure should not overwrite last-known-good actual cash data.
- Define the freshness tolerance for the external assumption feed.

19. Integration Monitoring Dashboard

Oracle-inspired | External Source Integration Monitor

External Data Reliability & Cash Forecast Cockpit



Recommended monitoring KPIs

- Source freshness by system.
- Rows/amount accepted vs rejected.
- Critical unmapped member count.
- Source-to-target variance.
- Latest successful job time and duration.
- Stale-source forecast exposure amount.
- Forecast impact from newly loaded external feeds.
- Retry queue / Integration Agent health where applicable.

20. Log and Error Examples

Sample External Data Integration Log

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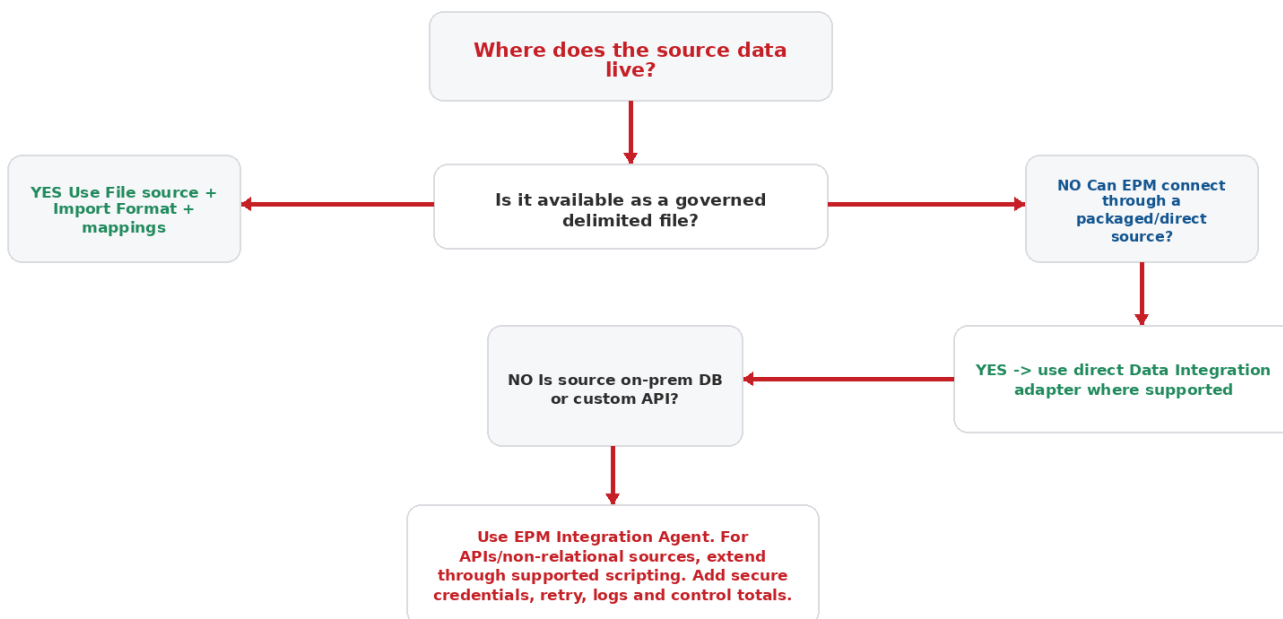
06:30:02 INFO PCF_TMS_13W_CASH started | W01-W13 | Mode=REPLACE
06:30:11 INFO Source file TMS_POSITION_20260826.csv | rows=12,480 | checksum verified
06:30:16 WARN 14 rows normalized: source week label 2026-W35 -> target W35
06:30:21 ERROR Currency GBPX not mapped | source row=8841 | Entity=UK01
06:30:24 ERROR OpeningCash line has invalid sign convention | row=9033
06:30:31 INFO Loaded rows=12,478 | rejected=2 | source total=18,450,000
06:31:03 WARN BANK_EU_02 source timestamp exceeds 4h freshness threshold
06:31:25 ERROR API_MACRO call timeout after 60s | retry=1/3 queued
06:32:10 INFO Integration ended PARTIAL_SUCCESS | publish blocked by Critical=2
  
```

Severity model

Severity	Example	Action
Critical	Invalid currency/entity, duplicate slice, source total mismatch	Block publish and remediate.
High	Stale bank/TMS source above SLA	Escalate; use approved last-known-good policy only if governance permits.
Medium	Non-material row rejected with known fallback	Document and track to closure.
Info	Normal row counts, checksum and job completion	Retain as audit evidence.

21. Troubleshooting Handbook

Choose the External Integration Pattern



Symptom	Likely cause	Remediation
File not parsed	Delimiter/header/encoding mismatch	Validate file options and sample preview; fix source contract.
All rows rejected	Wrong import-format dimension assignment	Correct source column-to-target dimension mapping.
Week shifted	Source/global period mapping incorrect	Correct explicit period mapping and rerun affected slice.
Duplicate values	Append/merge used incorrectly on rerun	Use idempotent replace strategy for controlled target POV.
Agent integration fails	Agent/cluster/network/credential issue	Check agent status, cluster assignment, logs and credential store.
API feed times out	External endpoint/network/extension script issue	Retry with backoff; preserve last-known-good; do not create partial data.
Cash sign reversed	Source convention inconsistent	Normalize sign with documented business rule and test inflow/outflow totals.
Source total != target	Rejects, rounding, mapping or filter mismatch	Reconcile by run ID and exception reason before publish.

22. Security, Credentials and Governance

Area	Design requirement
Location security	Grant users only the Data Integration locations they operate or review.
Service accounts	Use least privilege, named ownership and credential rotation.
Integration Agent	Harden host, restrict outbound/inbound paths per approved architecture, protect config/credential files.
File handling	Use controlled landing area, checksum, retention and duplicate detection.
API credentials	Keep secrets out of scripts/logs; use approved secret/credential handling.
Audit	Retain job IDs, user, source timestamp, mapping version, totals and approval status.
Change control	Promote mapping/query/script changes through DEV/TEST/PROD with evidence.

23. Automation and Production Runbook

Run phase	Example time	Owner	Control
Source cut-off	05:45	Source Ops	Source snapshot complete
Bank/TMS extract	06:00	Treasury Ops	Timestamp within SLA
Legacy/warehouse extract	06:10	Integration Ops	Counts/control totals captured
Data Integration loads	06:30	EPM Integration	Critical rejects = 0
PCF calculation	06:50	EPM	Rules successful
Reconciliation	07:05	Cash Analyst	Source-target difference = 0
Exception review	07:20	Cash Manager	Freshness/mapping issues cleared
Publish/approve	07:30	Treasury Manager	Workflow approved

Restart strategy

- Rerun only the affected source/location and period slice.
- Use the same business snapshot but a new technical job ID when retrying.
- Never append blindly after a partial load.
- After mapping changes, reprocess the complete affected target slice.
- Record last-known-good source timestamp and approved fallback behavior.
- Do not publish PARTIAL_SUCCESS when critical exceptions remain.

24. Advanced Student Challenge

1. Add a fourth external source from an on-prem relational treasury database using the EPM Integration Agent pattern.
2. Design an Integration Agent cluster and recovery plan for agent-node failure.
3. Create a custom external API feed for interest-rate assumptions through a governed staging pattern.
4. Build a W53 period-mapping test and prove no cash is lost or double counted.
5. Create one duplicate-file test and show how checksum/Run_ID logic blocks double loading.
6. Create a source-freshness rule that blocks publication when opening cash is older than one hour.
7. Demonstrate source-to-target reconciliation and explain the W6 liquidity shortfall to a treasury reviewer.

Assessment rubric

Area	Weight	Evidence expected
Architecture	20%	Correct pattern choice and separation of source/forecast responsibilities
Data Integration	25%	Source, import format, period and member mappings
Controls	25%	Freshness, idempotency, reconciliation and rejection handling
Forecast impact	20%	Correct closing cash and scenario decision
Operations	10%	Runbook, retry, security and support evidence

25. Implementation Checklist and Technical Reference

Area	Go-live requirement
Source inventory	Every external system classified by owner, grain, cut-off and integration pattern.
Target grain	Daily/weekly/monthly grain matches PCF feature enablement.
Templates	PCF sample data-load templates reviewed for target intersections.
Data Integration	Production integrations configured rather than relying on ad hoc imports.
Mappings	Entity, line item, currency, scenario and custom mappings approved.
Periods	Fiscal/ISO calendar and W53 edge cases tested.
Controls	Source totals, target totals, reject counts, checksums and timestamps captured.
Idempotency	Rerun strategy tested without duplicates.
Agent	Agent/cluster architecture tested where required.
Security	Location and service-account access is least privilege.
Forecast	Rules/data maps execute after successful data load.
Operations	Monitoring, alerts, fallback and incident owners documented.
UAT	Normal, stale, invalid, duplicate and partial-failure cases signed off.

Oracle reference notes

Current Oracle documentation states that PCF sample data import templates can be used as guidance for non-ERP data, while Data Integration is the preferred integration mechanism for external sources. Non-Fusion ERP data should be aggregated to the cash forecasting granularity enabled in the application. Data Integration supports file-based integrations and a range of direct source types; EPM Integration Agent supports on-prem relational sources and can be extended by scripting to reach additional sources including third-party REST APIs.

Training disclaimer

Oracle-inspired screens and diagrams in this guide are conceptual training designs, not pixel-for-pixel reproductions of Oracle product UI. Always validate release-specific navigation and supported connectors in the subscribed environment before production deployment.